Measurement: parameters, FLOPs, tokens, and loss

> **Reading note.** This is an explanatory reconstruction of Jiaxuan Zou's literature synthesis. Published experiments belong to the cited teams. Numbered protocols and illustrative matrices are our planning synthesis unless explicitly attributed.

A ladder fits numbers whose meanings must remain constant. The blog's §3 makes measurement a prerequisite to experiment design: changes in parameter conventions or token accounting can create apparent scaling effects.

Define the counters

Quantity	Definition to freeze	Log separately
N_{body}	Transformer trunk parameter count	Embeddings and output head
N_{active}	Parameters used per token in MoE	Total parameters and routing
D	Tokens contributing to the main loss	Processed and masked tokens
U	Deduplicated available unique tokens	Rewrites and repeated exposures
L	Held-out next-token cross-entropy	Auxiliary losses and task scores
C	Declared architecture/kernel FLOPs	Wall-clock time, communication, retries

For a standard multi-head-attention Transformer with a $4d$ feed-forward expansion and full attention-matrix accounting, the blog gives the approximation

$$C \approx \left(6 + \frac{\ell}{d}\right) N_{\text{body}} D + 6VdD.$$

Here ℓ is context length, d is hidden width, V is vocabulary size, and $N_{\text{body}} \approx 12d^2n$ for n layers. The final term represents the vocabulary projection. Causal kernels, grouped-query attention, gated MLPs, and MoE alter this accounting. Use the actual architecture and execution convention for decisions.

Procedure

1. Produce an exact parameter breakdown for every architecture configuration.
2. Record a FLOP accounting function and its assumptions.
3. Version tokenizer, packing, masks, loss reduction, and validation corpus.
4. Schedule evaluation at comparable fractions of training progress.
5. Log domain losses on a common evaluation distribution.

Training loss on different training mixtures changes with corpus entropy and is unsuitable as the sole mixture-ranking metric. Distributed averaging also requires a consistent denominator: averaging device means can bias loss when the devices have unequal numbers of valid tokens.

Deliverable: a measurement specification shared by every run, including units and exclusions.

Sources: blog §3; Porian et al., 2024, Table 2; Kaplan et al., 2020.

Blog/notes crosswalk · Detailed companion